

125A MIXENGINE

125A MixEngine - User Manual
Version 1.0.0 - Windows x64 VST3

125A AUDIO SOFTWARE

1. Overview

125A MixEngine is a modular coloration and summing processor delivered in two editions: **125A MixEngine** as a PreSonus Studio One Mix FX and **125A MixEngine Channel** as a normal VST3 insert. Both use the same central DSP core. Only the Mix FX edition can access neighbouring DAW channels and therefore provide true channel-to-channel crosstalk.

Signal flow: **INPUT -> CONSOLE -> TUBE -> TAPE -> GLUE -> VINYL -> STEREO -> OUTPUT.**

0 VU is a reference point, not a level that every musical signal must constantly hit. With dynamic material, the VU reading can be considerably lower than the DAW peak meter.

2. Installation and editions

- **125A MixEngine.vst3:** PreSonus Studio One Mix FX. Load it in the Mix FX slot of a bus/main section.
- **125A MixEngine Channel.vst3:** standard VST3 insert for tracks, buses and other VST3 hosts.
- Windows installation: copy the VST3 bundle to C:\Program Files\Common Files\VST3 and rescan plugins in the host.
- Both editions use separate plugin IDs and may be installed side by side.

3. Metering, Reference Level and gain staging

Reference Level defines which digital level corresponds to 0 VU: -18, -14 or -10 dBFS. The default is -14 dBFS.

The VU meter uses an energy envelope with roughly 300 ms settling behaviour. It is not a peak meter. The CLIP indicator triggers at peak ≥ 0 dBFS and holds for roughly 0.75 s.

VU Source Input is measured after Input Gain. **VU Source Output** is measured after the complete DSP including Output Gain.

Level Match is fixed parameter-dependent compensation. It is not an adaptive loudness normalizer.

4. Complete control reference

Bypass	Switch	Active / Bypass	Active	Bypasses coloration while preserving fixed host latency.
Input Gain	Knob	-12.0 to +12.0 dB	0.0 dB	Level before all coloration stages.
Reference Level	Selector	-18 / -14 / -10 dBFS	-14 dBFS	Defines 0 VU and internal calibration.
Level Match	Switch	Off / On	On	Fixed compensation for coloration stages.
Console On	Switch	Off / On	On	Enables Console.
Console Mode	Selector	Clean / Classic / Vintage / Modern	Classic	Four generic console transfer characters.
Console Drive	Knob	0 to 100 %	25 %	0 % is neutral in the nonlinear core; higher Drive adds saturation, peak rounding and density.
Crosstalk	Knob, Mix FX only	0 to 100 %	10 %	True bleed to direct neighbour channels; max nominal about 1.8 % (-34.9 dB) per neighbour.
Console Noise	Knob	0 to 100 %	0 %	Independent coloured console floor noise.
Tube On	Switch	Off / On	Off	Enables Tube.
Tube Voice	Selector	Soft / Balanced / Hot	Balanced	Selects progressively stronger drive/bias/asymmetry behaviour.
Tube Amount	Knob	0 to 100 %	20 %	Tube saturation amount; 0 % is neutral.
Tape On	Switch	Off / On	Off	Enables studio tape.
Tape Amount	Knob	0 to 100 %	20 %	Tape saturation, soft compression and wet contribution.
Tape Speed	Selector	7.5 / 15 / 30 ips	15 ips	Changes bandwidth, low bump and transport character.
Tape Stability	Knob	0 to 100 %	90 %	100 % = stable; lower values increase wow/flutter.
Tape Hiss	Knob	0 to 100 %	0 %	Independent tape-coloured hiss.

Glue On	Switch	Off / On	Off	Enables Glue.
Glue Amount	Knob	0 to 100 %	15 %	Increases compression strength and blend.
Glue Response	Knob	0 to 100 %	50 %	Low = slower/smoothier; high = faster/tighter.
Vinyl On	Switch	Off / On	Off	Enables Vinyl.
Vinyl Color	Knob	0 to 100 %	25 %	Base character: softer top end, body and saturation.
Vinyl Wear	Knob	0 to 100 %	0 %	Age/use: darker top end plus additional coloration and saturation.
Vinyl Surface	Knob	0 to 100 %	0 %	Surface noise plus occasional clicks/pops.
Depth	Knob	-100 to +100 %	0 %	Processes upper Side around 2 kHz and above; endpoints about +/-4 dB.
Width	Knob	0 to 200 %	100 %	M/S width; 100 % is neutral.
Low Mono	Knob	0 to 100 %	0 %	Progressively removes low Side with a fixed 120 Hz transition.
Output Gain	Knob	-12.0 to +12.0 dB	0.0 dB	Final level after all stages.
VU Source	Selector	Input / Output	Output	Chooses meter position.
Quality	Selector	Eco 1x / Normal 2x / High 4x	Normal 2x	Oversampling factor for relevant nonlinear stages.
UI Scale	Button	75 / 100 / 125 / 150 %	100 %	Each click cycles to the next GUI size.
VU L/R	Meter	-20 to +3 VU	-	Energy-based VU display.
CLIP L/R	Indicator	off / on	-	Peak >= 0 dBFS, hold about 0.75 s.

5. Console in detail

Drive 0 % is a true identity point in the nonlinear Console core. An enabled Console may still contain fixed properties such as channel tolerances, DC blocking, optional noise and - in Mix FX - Crosstalk. The selected mode's actual saturation increases with Drive.

- **Clean:** most restrained transfer.
- **Classic:** more density, slight low-frequency coupling and a small high-frequency/asymmetric component.
- **Vintage:** strongest low/asymmetric coloration and highest saturation tendency.
- **Modern:** more high-frequency-oriented shaping with a comparatively tight feel.

Crosstalk couples only **direct neighbours**. Channel 1 couples to channel 2, not directly to channel 3. A middle channel can couple to both direct neighbours. Coupling is lane-preserving: left stays left and right stays right. The standard Channel edition deliberately has no Crosstalk control.

6. Tube

Soft is the mildest voice, Balanced is the middle setting and Hot is the strongest. Amount 0 % is neutral. Increasing Amount raises drive, harmonic density and asymmetry.

7. Tape

Amount controls saturation and soft compression. **7.5 ips** rolls the top end off earlier and has the strongest low bump, **15 ips** is balanced, and **30 ips** is more extended and clean. Stability 100 % means stable transport; lower values increase wow/flutter. Hiss is independent and can remain at 0.

8. Glue

Glue is not a fully parameterised bus compressor. Amount increases the compression effect. Response moves internal attack/release behaviour from slower/smoothier toward faster/tighter. Stereo detection is linked.

9. Vinyl

Color sets the base character. **Wear** models age/use and can darken and saturate the signal without mandatory crackle. **Surface** is the actual surface layer with noise and click/pop events; Wear also affects their character when Surface is

active.

10. Stereo

Width is conventional M/S width control. 100 % is neutral.

Depth is deliberately not a second Width control. It only affects the upper Side region around 2 kHz and above. +100 % attenuates that region by about 4 dB; -100 % boosts it by about 4 dB.

Low Mono uses a fixed 120 Hz transition and progressively removes low-frequency Side content. 100 % is full Low Mono action, not a brickwall crossover.

11. Quality, latency and automation

Eco = 1x, Normal = 2x, High = 4x oversampling in relevant nonlinear stages.

The plugin reports a fixed host latency of **21 samples** independent of Quality mode. Shorter internal paths are aligned to that target and Bypass keeps the same latency.

Parameters are host-automatable. Processing uses the latest parameter value received for the current audio block; no claim of sample-accurate interpolation is made.

12. Technical data

- Format: VST3, Windows x64
- Editions: PreSonus Studio One Mix FX + Standard VST3 Channel
- Audio: mono/stereo, 32-bit and 64-bit sample processing
- Fixed host latency: 21 samples
- Oversampling: 1x / 2x / 4x
- 0 VU: -18 / -14 / -10 dBFS
- VU response: about 300 ms
- Clip hold: about 0.75 s
- Low Mono: fixed 120 Hz transition
- Depth: upper Side split around 2 kHz, endpoints about +/-4 dB
- Mix FX Crosstalk: direct neighbours, nominal max coefficient 0.018

13. Mastering note

125A MixEngine is a coloration processor, not a limiter or complete mastering system. Final loudness, true-peak limiting and platform targets are set with separate mastering tools.

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